MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 4J52A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Fixed frequency, 9,375 ±30 MHz, pulse type, 70-kw rated peak-power output, permanent magnet, air cooled.

ABSOLUTE RATINGS:

Parameter:	Ef	if	tk	Anode T	VSWR	Cathode bushing T
Unit:	V	a	sec	°C	Ratio	°C
Maximum:	14	10 (surge)	---	150	1.5:1	175
Minimum:	---	---	90	--- (see figure note t)	---	-55 (see figure note s)

ABSOLUTE RATINGS:

Parameter:	ib	Pi	tpc	rrv	Storage ambient T	Alt
Unit:	a	W	μs	kv/μs	°C	ft
Maximum:	15	240	5.5	187	+85	10,000
Minimum:	12	---	0.4	54	-55	---

PHYSICAL CHARACTERISTICS:

See figure 1
Weight: 6 pounds (approximate)

TEST CONDITIONS:

Parameter:	Ef	tpc	Du	rrv	Ib
Unit:	V	μs	---	kv/μs	mAdc
Tolerance:	--- (see note 5)	±10%	---	±10%	---
Test 1:	9.1	0.4	0.00065	170	9.8
Test 2:	9.1	0.4	0.00065	170	---
Test 3:	7.9	5.0	0.001	110	15
Test 4:	7.9	5.0	0.001	110	---
Test 5:	7.9	5.0	0.001	60	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST CONDITION	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	---					
1301	Heater current	---	---	Ef = 12.6 V; tk = 180 seconds (min)	If	2.0	2.4	A
4003	Pressurizing	---	---	40 to 45 psia, input and output assemblies	---	---	---	---
4250	Power output	3	1		Po	45	---	W
		3	3		Po	70	---	W
4308	RF bandwidth	6, 7, 17	2		BW	---	2.0/tpc	MHz
		6, 7, 18	4		BW	---	2.5/tpc	MHz
4308	Minor lobe ratio	6, 7, 17	2		Ratio	8.0	---	db
		6, 7, 18	4		Ratio	6.0	---	db
4306	Pulse voltage	---	3		epy	14	16	kv
4315	Stability	---	1		MP	---	0.25	%
		11	3		MP	---	0.25	%
4310	Frequency pulling factor	---	1		ΔF	---	13	MHz
---	Frequency	t	3	Anode temp at reference point = 100° ±10° C	F	9,345	9,405	MHz
4311	Frequency pushing factor	---	3	ib = 12 a to 15 a	ΔF	---	0.5	MHz/a
	Quality conformance inspection, part 2	2	---					
---	Vibration	15		Ef = 12.6 V; 15 G; F = 60 Hz; duration 15 minutes; no heater cathode shorts during test	---	---	---	---
---	Shelf life	8	---	t = 90 days	---	---	---	---
---	Shelf-life end points							
	Power output	---	3		Po	70	---	W
	Stability	---	3		MP	---	0.25	%
4315	Stability	---	3		MP	---	0.5	%
		---	5		MP	---	0.25	%
	Quality conformance inspection, part 3							
---	Life-test provisions							
1501	Intermittent life	12		Group D; VSWR = 1.5:1 (min) with phase varying through a minimum of 1/2 λ approximately every 15 minutes	---	833	---	Cycles
		16						

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METHOD	REQUIREMENT OR TEST	NOTES	TEST CONDITION	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued							
---	Life-test end points							
4250	Power output	3	3		Po	56	---	W
---	Frequency	---	3		F	9,345	9,405	MHz
4308	RF bandwidth	6, 7, 14	3		BW	---	2.5/tpc	MHz
4308	Minor lobe ratio	6, 7, 14	3		Ratio	6.0	---	db
4315	Stability	14 14	1 3		MP MP	--- ---	2.0 2.0	% %
	Periodic-check tests							
---	Vibration	15	1	10 G; F = 50 to 500 Hz; varied over complete range in not less than 5 minutes at uniform rate	ΔF	---	±0.5	MHz
---	Shock	4	---	50 G; 4 ms duration; no voltages	---	---	---	---
---	Input capacitance	---	---	Measured between cathode terminal and mounting plate	Cin	10	14	pF
4309	Phase of sink	9	---		Distance	0.26	0.4	λg
4027	Temperature coefficient	t	1	Anode temp = 70° to 100° C at reference point	ΔF/°C	---	0.25	MHz
---	Low-temperature operation (stability)	---	1	tk = 90 sec max	MP	---	1.0	%
1143	Forced convection	10	1		ΔT	---	50	°C
---	Low-pressure operation	13	3	Pressure = 525 mm Hg absolute	---	---	---	---

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 0.65 percent, inspection level II.
2. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 2, shall be 6.5 percent, inspection level S3.
3. The tube heater shall be protected against arcing by the use of a capacitor that places a minimum capacitance of 4,000 pF across the heater directly at the input terminals.

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NOTES: -Continued

4. (a) This test shall be performed on the Naval Research Laboratory's standard shock machine for electronic devices, or approved equivalent. A resilient cushion (see note 4(b)) shall be interposed between hammer and anvil of table and a suitable hammer angle selected to produce a shock of the specified magnitude and duration (see note 4(c)). The mounting plate of the tube shall be bolted with brass bolts to either the table or the standard angle bracket, depending upon the direction of the desired shock, using a 1-9/16-inch thick brass spacer between the tube mounting plate and the table or angle bracket. The shock shall be measured on the brass spacer. The tube shall be given one shock in each of the following directions:

   (1) Parallel to cathode, with cathode terminals pointing away from the hammer.

   (2) Perpendicular to cathode axis and output waveguide axis.

   (3) Perpendicular to cathode axis and parallel to the output waveguide axis.

   (b) A resilient cushion consisting of 9/32-inch thick rubber sheet of 30-Shore Durometer hardness, covering the entire anvil of the table, has been found to produce the specified shock duration under the given conditions of table load and shock magnitude.

   (c) Because of the varying resilience of the tube on its mounting plate, with different shock directions, and the high ratio of tube to table weight, the hammer angle will vary with the tube orientation to produce the required magnitude of shock.

   (d) Criteria for passing shock test: After the shock test, the tube shall show no mechanical failure and shall meet all electrical requirements of the tube specification sheet with the exception of life tests.

5. A tk of 180 seconds (maximum) with an Ef of 12.6 volts is required before test.

6. The tube shall be operated into a transmission line with a VSWR of 1.5:1 adjusted in phase to produce maximum spectrum degradation.

7. A suitable spectrum shall be considered one in which the major lobe has a shape such that its slope does not change sign more than once for power levels greater than the specified db below its peak.

8. This test shall be performed on four tubes per month when tube is in continuous production, but shipments of that month's production shall not be held pending completion of the test. So long as three of the four tubes for each of the first 3 months of a production run pass the test, and 75 percent of the cumulative quantity of tubes tested pass the test, tubes shall be considered to conform to this specification. If either of the conditions are not met, shipments shall be halted until three of four tubes of current production conform to test.

9. Using a standard cold-test technique, the phase of sink as measured from the output flange to the first minimum, toward the load, shall be within the limits specified.

10. An airflow of 15 cfm at approximately 760 mm Hg shall be directed on the cooling fins from an orifice of 2-1/2 by 1-3/16 inches. The temperature rise shall be measured at that point on the anode block as shown on figure 1.

11. This test shall be the first one performed after the specified holding period.

12. Air cooling shall be adjusted so that the anode block runs at 150° C, or at the maximum temperature it will reach in the absence of cooling, whichever is lower. This shall be adjusted during the test condition 3 portion of the cycle.

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NOTES: -Continued

13. The tube shall be operated in a transmission line with a load VSWR of 1.5:1 and variable phase. At specified pressure, there shall be no evidence of breakdown at the input or output assemblies.

14. If during life test the tube does not meet the specified limits, it shall be recycled for an additional five cycles. At such time, the tests shall be repeated. Should the tube fail the second test, it shall be considered unsatisfactory.

15. Direction of vibration shall be in a plane perpendicular to the axis of the cathode.

16. Life test shall be conducted in accordance with the following cycle:

Condition	Ib	Ef	Duration
Standby	---	12.6 V	3 minutes
Test 1	9.8 mA	9.1 V	3 minutes
Test 3	15.0 mA	7.9 V	15 minutes
Off	---	0	9 minutes (minimum)

17. Measurements shall be made at 11 a and 15 a.

18. Measurements shall be made at 12 a and 15 a.

Custodians:
Army - EL
Navy - EC
Air Force - 85

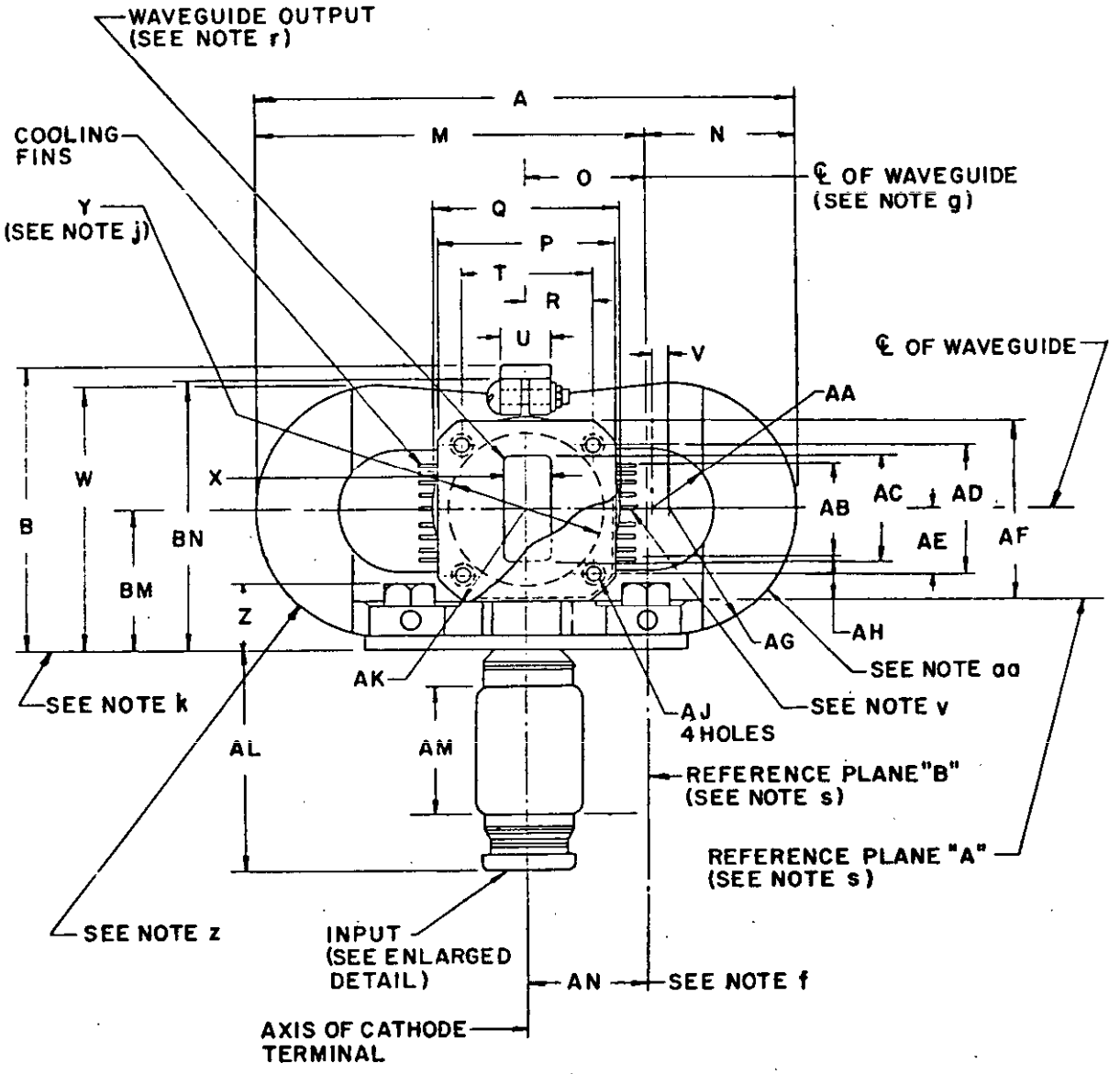
Review activities:
Army - EL
Navy -
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, CG, SH
Air Force - 19

Preparing activity:
Navy - EC

(Project 5960-2343)

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FRONT VIEW

FIGURE 1. Outline drawing of electron tube type 4J52A.

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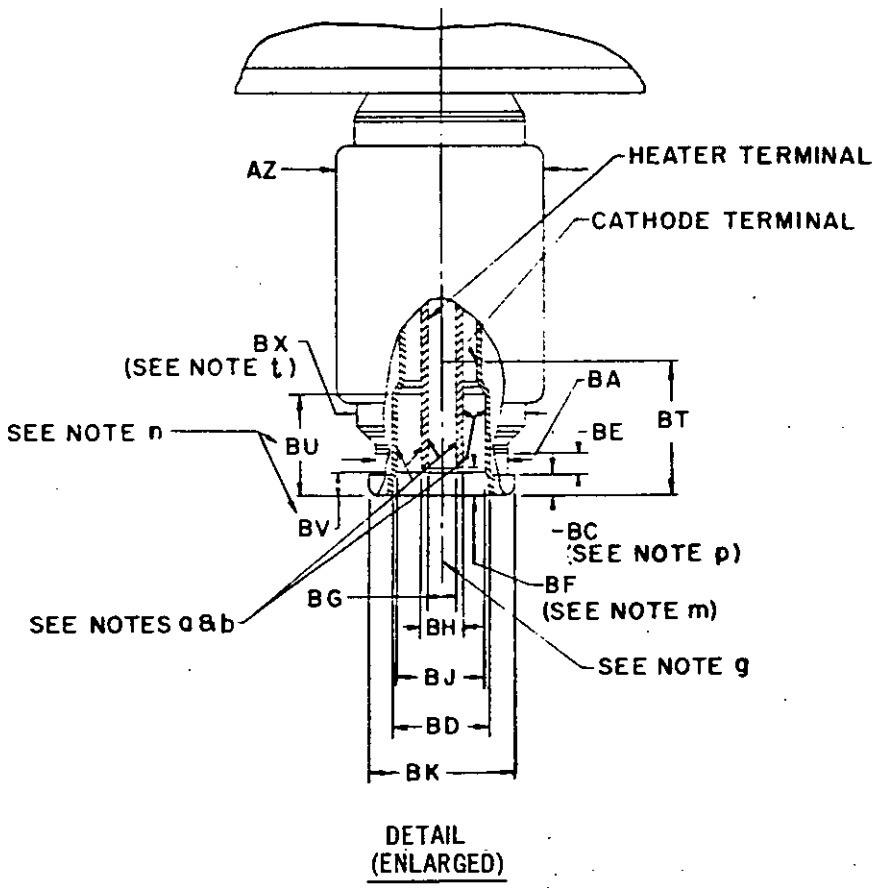


FIGURE 1. Outline drawing of electron tube type 4J52A - Continued.

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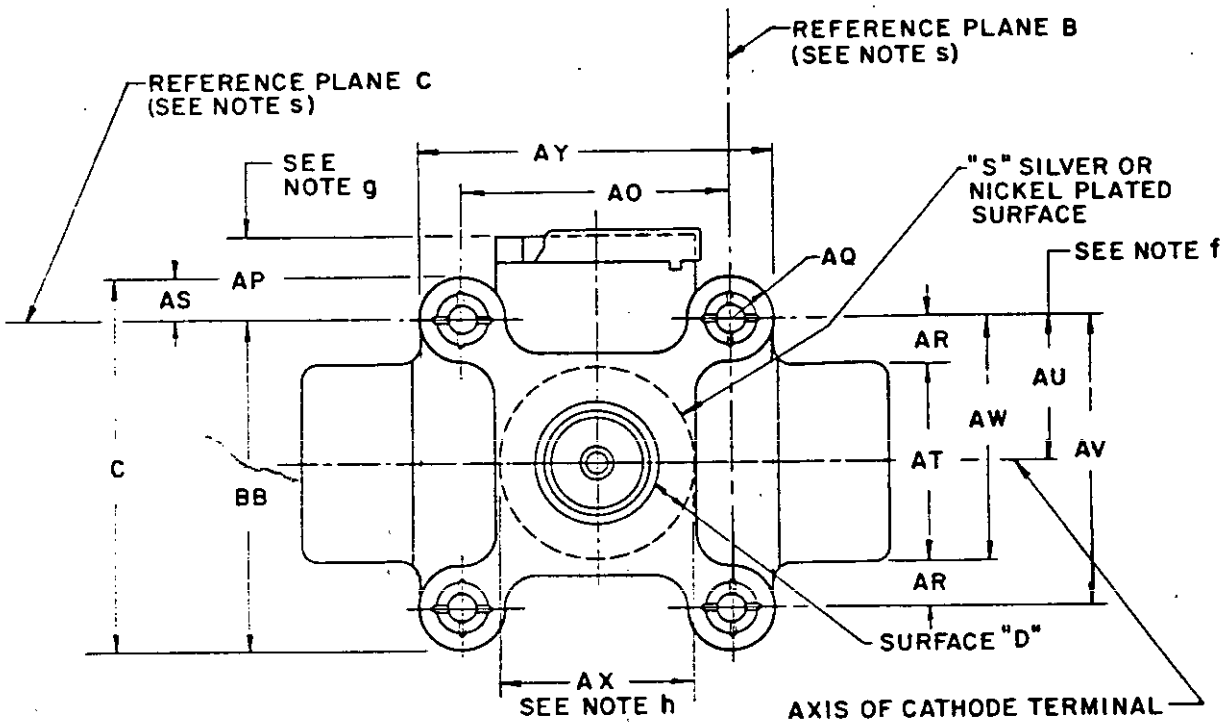
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 1			a
O	1.250 (31.75)	1.280 (32.51)	g
R	.732 (18.59)	.742 (18.85)	
T	1.470 (37.34)	1.478 (37.54)	
Z	.594 (15.09)	.656 (16.66)	
AD	1.348 (34.24)	1.356 (34.44)	
AE	.671 (17.04)	.681 (17.30)	
AL	2.094 (53.19)	2.219 (56.36)	
BC	.115 (2.92)	.135 (3.43)	p
BD	.610 (15.49)	.650 (16.51)	
BF	.125 (3.18)	.187 (4.75)	m
BG	.164 (4.17)	.174 (4.42)	
BH	.234 (5.94)	.266 (6.76)	
BJ	.532 (13.51)	.545 (13.84)	
BK	.825 (20.96)	.838(21.29)	
BM	1.386 (35.20)	1.426 (36.22)	
Quality conformance inspection, part 2			b
A	- - -	5.938(150.83)	
B	- - -	3.156 (80.16)	
M	- - -	4.250(107.95)	
N	- - -	1.688 (42.88)	
Q	- - -	1.906 (48.41)	
U	- - -	1.438 (36.53)	
W	- - -	2.844 (72.24)	
AM	1.250 (31.75)	- - -	
BE	.125 (3.18)	- - -	
BN	- - -	2.938 (74.63)	
BT	.750 (19.05)	- - -	
BU	.516 (13.11)	- - -	n
BV	- - -	.200 (5.08)	n

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	Notes
Quality conformance inspection, part 3 (periodic check)		c
	Nominal	
P	1.830 (46.48)	
V	.141 (3.58)	j
X	.497 (12.62)	
Y	1.625 (41.28)	
AA	.625 (15.88)	
AB	.984 (24.99)	
AC	1.122 (28.50)	
AF	1.830 (46.48)	
AG	1.328 (33.73)	
AH	.922 (23.42)	
AJ	No. 8-32 UNC-2B	
AK	1.156 (29.36)	
AN	1.265 (32.13)	f
AZ	1.125 (28.58)	
BA	.750 (19.05)	
BX	1.000 (25.40)	t

FRONT VIEW & DETAIL DIM

FIGURE 1. Outline drawing of electron tube type 4J52A - Continued.

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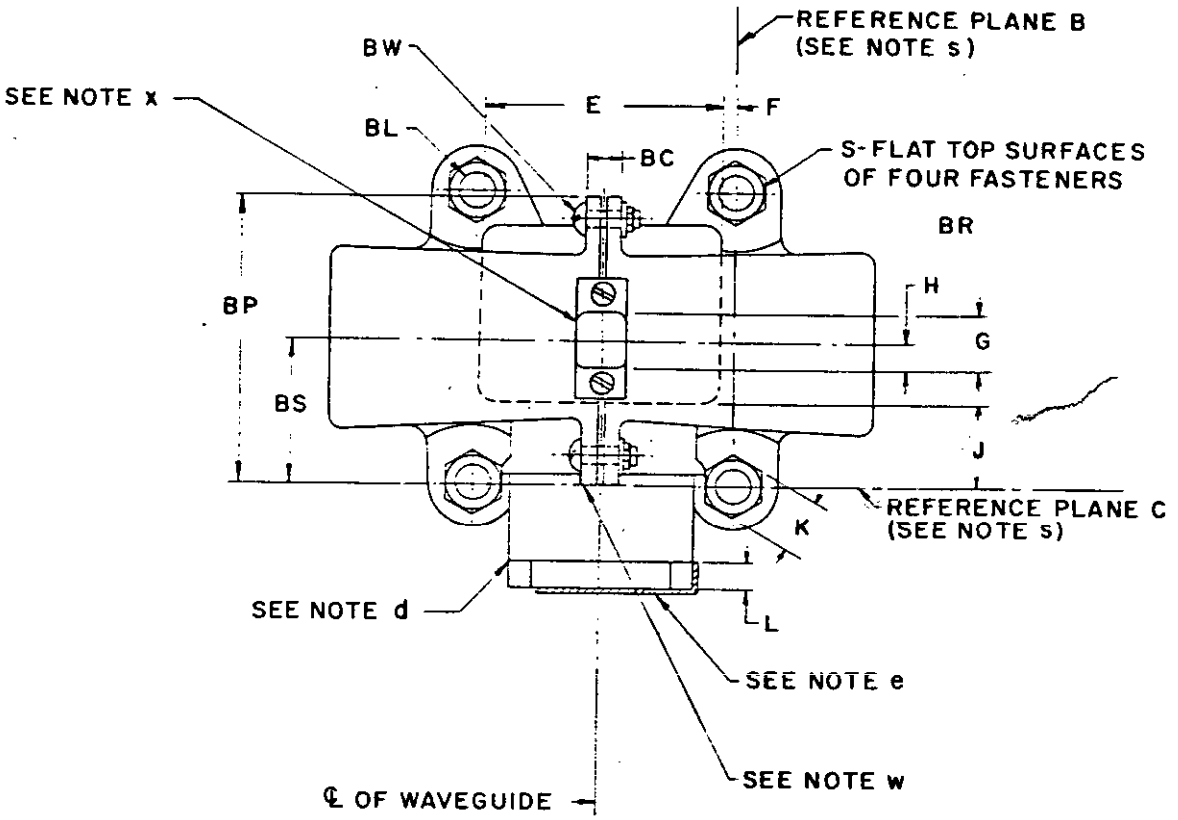


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 1			a
AP	.990 (25.15)	1.040 (26.42)	g
AV	2.771 (70.38)	2.791 (70.89)	
AO	2.521 (64.03)	2.541 (64.54)	
Quality conformance inspection, part 2			b
C	- - -	3.625 (92.08)	
AR	- - -	.594 (15.09)	
AS	- - -	.422 (10.72)	
AT	- - -	1.938 (49.23)	
AW	- - -	2.375 (60.33)	
AX	1.875 (47.63)	- - -	h
AY	- - -	3.438 (87.33)	
BB	- - -	3.203 (81.36)	
BP	- - -	2.875 (73.03)	
Quality conformance inspection, part 3 (periodic check)			c
	Nominal		
AQ	.375 (9.53)		
AU	1.391 (35.33)		f

BOTTOM VIEW

FIGURE 1. Outline drawing of electron tube type 4J52A - Continued.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 1			a
BS	1.344 (34.14)	1.438 (36.53)	
BL	.276 (7.01)	.286 (7.26)	
Quality conformance inspection, part 2			b
G	- - -	.812 (20.62)	
H	- - -	.406 (10.31)	
BQ	- - -	.625 (15.88)	
BR	.031R (.79)		
Quality conformance inspection, part 3 (Periodic check)			c
	Nominal		
E	2.375 (60.33)		
F	.078 (1.98)		
J	.797 (20.24)		
K	.500 (12.70)		
L	.250 (6.35)		
BW	No. 6-32 UNC-2A		

TOP VIEW

FIGURE 1. Outline drawing of electron tube type 4J52A - Continued.

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NOTES:

a. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1 shall be 1 percent with inspection level of II.
b. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 2 shall be 6.5 percent with inspection level of S3.
c. If these tests have been previously performed by the tube manufacturer, and if the materiel, manufacturing process and test requirements are the same as when the tests were performed, the subject test(s) need not be performed.
d. All metal surfaces black finished except those designated surface S and surface D. Surfaces S shall be gold, silver, nickel or bright alloy plated, or brass surfaces, except as specified.
e. Hermetic connections may be made to surface D.
f. Axis of cathode terminal shall occupy specified location within .047 (1.19 mm) radius. Note g shall apply.
g. Limits include angular and lateral deviations.
h. With diameter AX resting on a plane surface coincident with reference plane A, a gage .010 (.25 mm) thick and .125 (3.18 mm) wide shall not enter, and areas of the base plate outside diameter AX shall be within .010 (.25 mm) of the plane surface.
j. With diameter Y resting on a plane surface, a gage .005 (.13 mm) thick and .125 (3.18 mm) wide shall not enter.
k. Any portion of the assembly which extends below reference plane A shall be within .625 (15.88 mm) radius of specified axis of input.
m. Defines extremity of cylindrical section designated BG.
n. Defines extremity of cylindrical section designated BJ.
p. Clamping shall be confined to this area.
q. Heater terminal and cathode terminal shall be concentric within .010 (.25 mm).
r. Protective covers shall be provided with tube.
s. Reference planes A, B and C are mutually perpendicular.
t. Reference point for cathode temperature measurements. Locate on dimension BX.
u. Gold, silver or nickel plate over full extent of maximum plug penetration.
v. Reference point for anode temperature measurements. Locate where a central fin meets the anode block near the output section.
w. Extremity of magnet lug shall coincide with reference plane C within .094 (2.39 mm).
x. Seal-off protector may be round in shape.
y. For vibration tests, planes of vibration shall be reference planes A, B and C.
z. See note aa. Warning: Maintain minimum 2 inches (50.8 mm) between this magnet and magnetic materials (magnets, steel tool, plates, etc).
aa. Manufacturers code designation and tube serial number shall appear within this area.
ab. Heater and cathode terminals shall be gold plated over the full plug connection areas in accordance with provisions of MIL-G-45204, type 1, class 1.

FIGURE 1. Outline drawing of electron tube type 4J52A - Continued.

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